

Lab Prerequisites

Ubuntu Setup for Kubernetes Labs

Supported OS

- Ubuntu 20.04 LTS
 - Ubuntu 22.04 LTS
 - Ubuntu 24.04 LTS
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What We Will Install

By the end of this setup, you will have:

- Docker (container runtime)
- kubectl (Kubernetes CLI)
- Minikube (local Kubernetes cluster)
- Helm (package manager for Kubernetes)

These are the **only prerequisites** needed for the lab.

Step 1: Update the System

```
sudo apt update && sudo apt upgrade -y
```

Step 2: Install Required Base Packages

```
sudo apt install -y \  
  ca-certificates \  
  curl \  
  gnupg \  

```

```
lsb-release \
apt-transport-https
```

Step 3: Install Docker

Add Docker's official GPG key

```
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
```

Add Docker repository

```
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/
keyrings/docker.gpg] \
  https://download.docker.com/linux/ubuntu \
  $(lsb_release -cs) stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

Install Docker

```
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io
```

Enable Docker and add your user

```
sudo systemctl enable docker
sudo systemctl start docker
sudo usermod -aG docker $USER
```

👉 **Log out and log back in** (important).

Verify:

```
docker ps
```

You should not see permission errors.

Step 4: Install kubectl

```
curl -LO "https://dl.k8s.io/release/$(curl -Ls https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"  
chmod +x kubectl  
sudo mv kubectl /usr/local/bin/
```

Verify:

```
kubectl version --client
```

Step 5: Install Minikube

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64  
sudo install minikube-linux-amd64 /usr/local/bin/minikube
```

Verify:

```
minikube version
```

Step 6: Start Minikube (Docker Driver)

Minikube will use Docker as its runtime.

```
minikube start --driver=docker
```

Verify cluster:

```
kubectl get nodes
```

You should see **1 node in Ready state**.

Step 7: Install Helm

```
curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
```

Verify:

```
helm version
```

Final Verification Checklist

Run all of these:

```
docker ps
kubectl get nodes
helm version
minikube status
```

If all commands work, your system is **ready for the lab**.

Common Issues & Fixes

 **docker: permission denied**

Fix:

```
sudo usermod -aG docker $USER
logout
```

Minikube fails to start

Make sure Docker is running:

```
sudo systemctl status docker
```

Restart if needed:

```
sudo systemctl restart docker
```
